# Communication Lower Bounds for Multi-Party Graph Traversal

## Quarter 2 Project Report

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**Abstract**

A large number of computational problems can be modelled as communication games. For example, a graph traversal problem can be modelled as follows: Two players, Alice and Bob, have subgraphs of a known graph $G$, and wish to find a path from vertex $v$ to $w$ using only vertices and edges in either player's subgraphs. Communication complexity studies such situations by creating asymptotic bounds on the minimum amount of communication necessary to complete these tasks. In this report, we prove communication lower and upper bounds for some graph traversal problems, such as the example above.

Website: https://rybplayer.github.io/capstone/
Repository: https://github.com/rybplayer/DSCCapstone

# 1 Introduction

## 1.1 Motivating Communication Complexity

As networks and data grow in complexity and size, it will become increasingly difficult for individual machines to compute functions on such data. For example, suppose we wish to fly from San Diego to Santorini. There is no single airline that has both flights out of San Diego or into Santorini. Therefore, we will need to consult the flight database of various airlines to construct such a route. Do we need to ask every airline provider to communicate their entire schedule of flights to us? Are there techniques that we can employ to reduce the amount of flights we have to search through?

Bounding the amount of communication necessary to solve such problems is the primary objective of communication complexity. By studying communication complexity, we find lower and upper bounds on the minimum amount of communication necessary to solve such problems. In practice, these bounds can inform us whether certain algorithms are optimal, or if it is possible to optimize them further.

## 1.2 Communication Complexity Basics

Communication complexity studies the communication requirements to solve problems where inputs are distributed among at least two parties, each with unbounded computing power. The two-party problem is typically stated as follows:

> There are two players, Alice and Bob, who respectively have a $n$-bit string, $a$ and $b$, that may or may not be the same. The goal is to compute a boolean function, $f(a,b)$, that depends on both $a$ and $b$. It suffices for one player to find the answer $f(a,b)$, since communicating this to the other player can always be done in constant communication.

Under the two party model, it is always possible to compute $f(a,b)$ by sending all of $a$ to $b$, since we assume each party has unbounded computing power. This approach of sending the entire input from one party to another is what is called the *trivial solution*. We say problems are *maximally hard* or *maximally complicated* when we cannot do asymptotically better than the *trivial solution*.

We call such problems maximally hard because they have a deterministic communication complexity of $\Theta(n)$ bits. We call it *deterministic* as we do not allow error. Thus in maximally hard problems, Alice cannot do asymptotically better than sending the $n$ bits of $a$ to Bob, and likewise for Bob. An example of a maximally hard problem is 2-player equality ($\mathsf{EQ}$):

**Claim 1.** *Let $a,b \in \{0,1\}^n$. That is, $a$ and $b$ are $n$-bit strings. Define equality as*

$$\mathsf{EQ}(a,b) = \begin{cases} 1 & a = b, \\ 0 & a \neq b. \end{cases} \tag{1}$$

*The deterministic communication complexity of two party equality, denoted $D(\mathsf{EQ})$, is $\Omega(n)$.*

*Proof.* We can represent all possible $n$ bit inputs to the two parties using a $2^n \times 2^n$ boolean matrix $M$, whose columns are indexed by $a$ and whose rows by $b$. Then, let $M_{b,a} = 1$ if $a = b$ and $M_{b,a} = 0$ otherwise.

Therefore, $M$ is the identity matrix. It is a well-known fact that the deterministic communication complexity of a problem is at least the $\log_2$ rank of its boolean matrix (Rao and Yehudayoff 2020). Therefore, since equality's matrix $M$ is the $2^n \times 2^n$ identity matrix with rank $2^n$, we have $D(\mathsf{EQ}) = \Omega(\log_2 2^n) = \Omega(n)$. $\square$

$$M_{b,a} = \begin{array}{c|cccc} b\backslash a & 00 & 01 & 10 & 11 \\ \hline 00 & 1 & 0 & 0 & 0 \\ 01 & 0 & 1 & 0 & 0 \\ 10 & 0 & 0 & 1 & 0 \\ 11 & 0 & 0 & 0 & 1 \end{array}$$

**Figure 1:** *Communication matrix for the equality function on two-bit inputs. The entry at column $a$ and row $b$ is 1 when $a = b$ and 0 otherwise.*

As we will soon see, problems like equality are surprisingly useful for proving communication lower bounds for other, more complicated problems. We can do this by *embedding* equality into a more complex problem, such that an efficient solution to the more complicated problem can also efficiently solve equality. More generally, we say problem $A$ is *at least as hard* as problem $B$ if we can embed $B$ in $A$. The communication complexity of $A$ is then at least the communication complexity of $B$.

## 1.3 The Graph Traversal Problem

In this report, we study the problem of finding a path in a graph whose vertices may be owned by various players. This is an abstract version of the motivating problem in 1.1, which we choose to work in so that that any bounds we prove will be applicable to a wide range of problems.

Informally, the graph traversal problem presents a graph, where players may own vertices. There is a start and end vertex. The players wish to know whether there exists a path from the start to the end through only vertices owned by some player. Each player knows what the graph looks like, and which vertices they own though not which vertices are owned by other players.

Formally, the graph traversal problem for two players, denoted $\mathsf{TRAV}_2$, is defined as follows:

**Definition 1.** *Let $G = (V, E)$ be an unweighted, undirected graph, where $V$ is the set of all nodes and $E$ is the set of edges. Alice and Bob respectively own a vertex subset $V_A, V_B \subseteq V$ of the* *graph, as well as all edges $E_A = \{(u,v) \in E \mid \{u,v\} \cap V_A \neq \emptyset\}$ and similarly for $E_B$. These are all edges connected to some vertex in $V_A$ or $V_B$, respectively. Thus each player respectively owns a subgraphs $G_A = (V_A, E_A)$, $G_B = (V_B, E_B)$ of $G$. Note that multiple players can own the same vertices and edges, and some vertices and edges may have no owner whatsoever.*

*Let the graph $G$ have $|V| = n$ vertices. Then each player's subgraph can be represented by $n$-bit strings $x$ and $y$, where 1 at index $j$ indicates the player owns vertex $j$, and 0 means the player does not own $j$. This definition of $x$ and $y$ mean that we can use $x$ and $y$ interchangeably with $G_A$ and $G_B$ respectively. Now, given a fixed $G$, start vertex $v$, and end vertex $w$, an instance of* $\mathsf{TRAV}_2$ *is defined as follows:*

$$\mathsf{TRAV}_2(x,y) : \{0,1\}^n \times \{0,1\}^n \to \{0,1\} \tag{2}$$

$$\mathsf{TRAV}_2(x,y) = \begin{cases} 1 & \exists \textit{ path from } v \textit{ to } w \textit{ through only vertices and edges in } G_A \cup G_B. \\ 0 & \textit{There is no such path.} \end{cases} \tag{3}$$

The following figures illustrate what $\mathsf{TRAV}_2$ may look like.

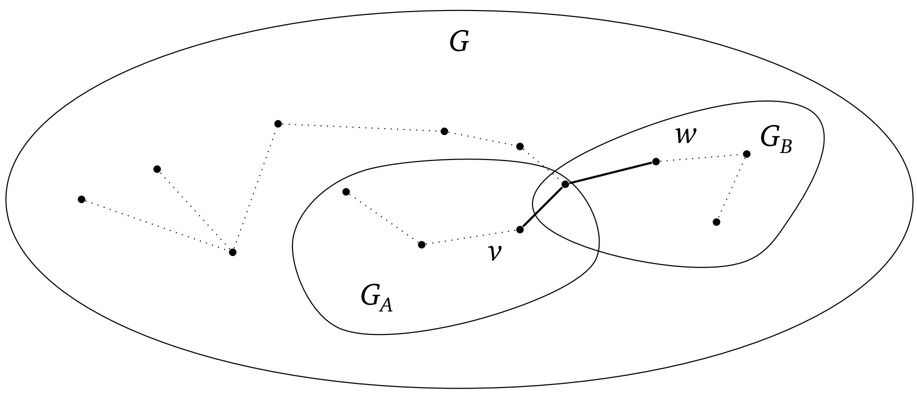


**Figure 2:** *Example graph where* $\mathsf{TRAV}(x,y) = 1$.

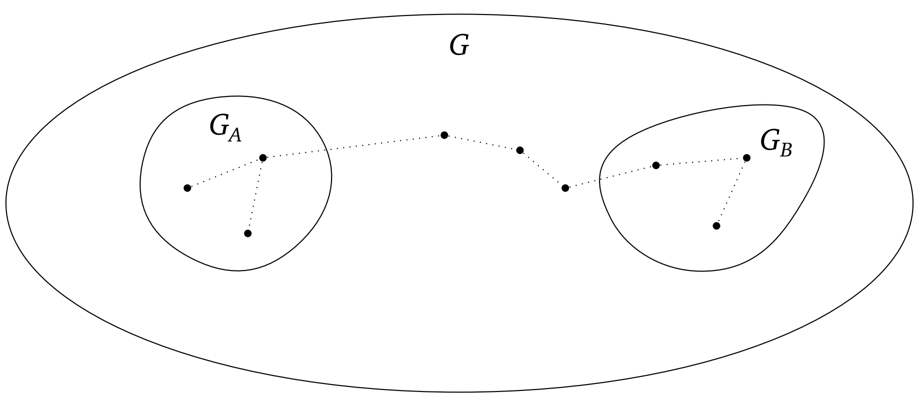


**Figure 3:** *Example graph where* $\mathsf{TRAV}(x,y) = 0$.

## 2 The Two-Party Traversal Problem

### 2.1 Deterministic Two-party Bounds

We claim that $\mathsf{TRAV}_2$ is at least as hard as two player equality:

**Claim 2.** *Recall that equality* $\mathsf{EQ}$ *has inputs* $x, y \in \{0,1\}^n$, *and* $\mathsf{EQ}(x, y) = 1$ *iff* $x = y$. *Then* $D(\mathsf{TRAV}_2) \geq D(\mathsf{EQ})$, *and more generally* $\mathsf{TRAV}_2$ *is at least as as hard as* $\mathsf{EQ}$.

The high-level idea of the following proof is to construct an equality gadget that lets us embed $\mathsf{EQ}$ into $\mathsf{TRAV}_2$. That is, given $x$ and $y$ as inputs of $\mathsf{EQ}$, we can convert these into subgraphs $G_1$ and $G_2$ inputs of $\mathsf{TRAV}$. We convert this in such a way that $\mathsf{TRAV}(G_1, G_2) = 1$ if and only if $\mathsf{EQ}(x, y) = 1$. Then $\mathsf{TRAV}$ cannot use less communication than the optimal $\mathsf{EQ}$ protocol, else we get a contradiction.

*Proof.* **Equality gadget.** We construct a gadget that enforces equality at a single bit position at index $i$ of $x$ and $y$, for $i \in [n]$. The gadget is shown in Figure 4, and consists of two parallel branches:

1. The top branch corresponds to the assignment $x_i = y_i = 1$,
2. The bottom branch corresponds to the assignment $x_i = y_i = 0$.

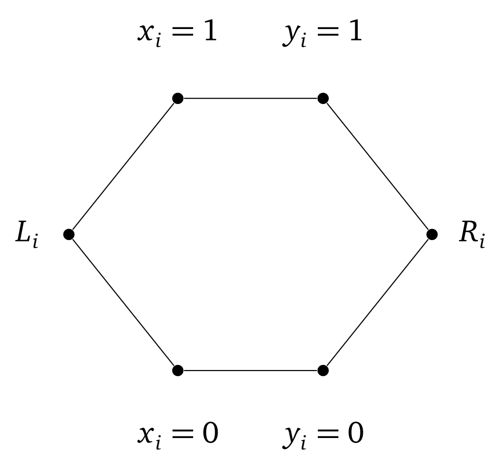


**Figure 4:** *Gadget for proving equality. A path only exists if $x_i = y_i$.*

Add $L_i$ and $R_i$ into $G_A$ and $G_B$. Then, add ownership of $x_i = 1$ to $G_A$ if $x_i = 1$ is true, else add $x_i = 0$. Do the same for Bob and $y_i$. Then a path from $L_i$ to $R_i$ exists if and only if Alice and Bob have vertices on the same branch, which holds exactly when $x_i = y_i$.

**Embedding** $\mathsf{EQ}$ **into** $\mathsf{TRAV}_2$**.** Construct a graph of $n$ equality gadgets in a chain, as in Figure 5. We do this by setting $R_l = L_{l+1}$ for $l \in [n-1]$. Construct subgraph $G_A$ as follows: Include all vertices $L_i$ and $R_i$ for $i \in [n]$. Also, for every $i \in [n]$, give $G_A$ ownership of the vertex labelled $x_i = 1$ if this is true, else $x_i = 0$. Repeat for $G_B$ and $y$.

In the resulting graph, set $v = L_1$ as the start vertex $v$, and $w = R_n$ as the end vertex. Then there exists a path from $v$ to $w$ if and only if all gadgets are traversable, which occurs if and only if $x_i = y_i$ for all $i \in [n]$, and therefore if and only if $x = y$.

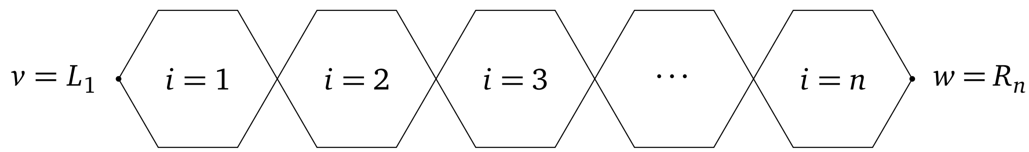


**Figure 5:** *$n$ equality gadgets chained together.*

We conclude by noting that the resulting graph $G$ has $\Theta(n)$ vertices and $\Theta(n)$ edges. As $|x| = |y| = \Theta(n)$, this completes the reduction and $\mathsf{TRAV}_2$ is at least as hard as $\mathsf{EQ}$. $\square$

The trivial protocol where Alice sends all of $G_A$ to Bob has communication complexity $\Theta(n)$. Therefore, $D(\mathsf{TRAV}_2) = O(n)$. Together with Claim 2, this gives $D(\mathsf{TRAV}_2) = \Theta(n)$. Thus $\mathsf{TRAV}_2$ is deterministically maximally hard.

## 2.2 Randomized Two-Party Bounds

Above, we note that the problem is *deterministically maximally hard*. In many cases, allowing for some error may allow us to produce a randomized algorithm with lower communication complexity.

Consider the *public coin randomness* model. In this model, both parties have access to the same random bit-string in addition to the problem description and input. This random bit string can be as long as we want. We also allow our output to error, that is, be incorrect, some bounded constant percent of the time, and strictly less than half the time. It can be shown that using the public coin randomness model, equality—which is Deterministically maximally hard—can be reduced to being $O(1)$. We can denote the randomized communication complexity of equality by $R(\mathsf{EQ}) = O(1)$.

Unfortunately, for $\mathsf{TRAV}_2$ we find that such a bound is unattainable. To prove this, we introduce the disjointness problem $\mathsf{DISJ}$, which has public coin randomized communication complexity $R(\mathsf{DISJ}) = \Theta(n)$. As we show, $\mathsf{DISJ}$ can be embedded into $\mathsf{TRAV}_2$.

**Definition 2.** *Define the disjointness problem as follows: Let $x, y \in \{0,1\}^n$ be $n$-bit strings representing subsets of $[n]$. Then, for all $i \in [n]$, if $x_i = 1$, we say $i$ is in Alice's subset $S_A \subseteq [n]$, and if $x_i = 0$ it is not. Similarly for $y$ and Bob. Then*

$$\mathsf{DISJ}(x, y) : \{0,1\}^n \times \{0,1\}^n \to \{0,1\} \tag{4}$$

$$\mathsf{DISJ}(x, y) = \begin{cases} 1 & S_A \cap S_B = \emptyset, \\ 0 & S_A \cap S_B \neq \emptyset. \end{cases} \tag{5}$$

*It is well-known that $D(\mathsf{DISJ}) = \Theta(n)$ and $R(\mathsf{DISJ}) = \Theta(n)$ (Rao and Yehudayoff 2020).*

**Claim 3.** *We claim $R(\mathsf{TRAV}_2) \geq R(\mathsf{DISJ})$, and more generally $\mathsf{TRAV}_2$ is at least as hard as $\mathsf{DISJ}$.*

*Proof.* **Disjointness gadget.** We construct a gadget that enforces disjointness at a single entry $i \in [n]$. The gadget is shown in Figure 6. There, let $x_i$ and $y_i$ be indicator functions that equal 1, respectively, when $i \in S_A$ and $i \in S_B$, else equal 0. Then:

1. The top branch corresponds to when $i \in S_A$ but $i \notin S_B$.
2. The middle branch corresponds to when $i \notin S_A$ and $i \notin S_B$.
3. The bottom branch corresponds to when $i \in S_B$ but $i \notin S_A$.

Give $L_i$ and $R_i$ to both $G_A$ and $G_B$. Then give ownership of vertices labelled $x_i = 1$ to $G_A$ if this is true, else give all vertices labelled $x_i = 0$. Do the same with $y$ and $G_B$. Therefore, there is no path $L_i$ to $R_i$ if $i \in X \cap Y$, but there is a path in all other cases.

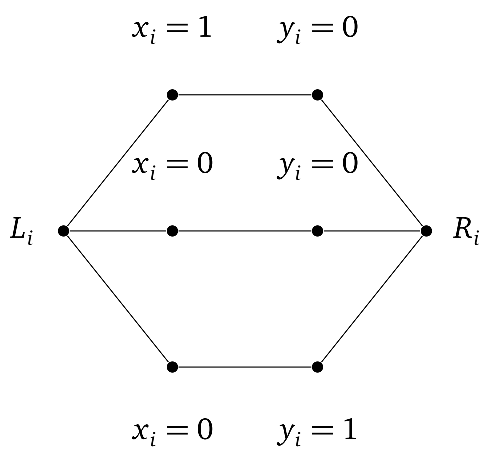


**Figure 6:** *Disjointness gadget. A path $L_i$ to $R_i$ exists only if $x_i = 0$ or $y_i = 0$.*

**Embedding** DISJ **into** $\mathsf{TRAV}_2$**.** We construct a graph $G$ by chaining $n$ copies of the disjointness gadget. More precisely, set $R_l = L_{l+1}$ for $l \in [n-1]$. Then, set $v = L_1$ as the start vertex and $w = R_n$ as the end vertex. Construct $G_A$ by giving ownership of all vertices $L_i$, $R_i$ and vertices labelled $x_i = 1$ if it is true, else $x_i = 0$, for $i \in [n]$. Do similar for $G_B$.

In the resulting graph, there exists a path from $v$ to $w$ if and only if all disjointness gadgets are traversable, which occurs if and only if $X \cap Y = \emptyset$.

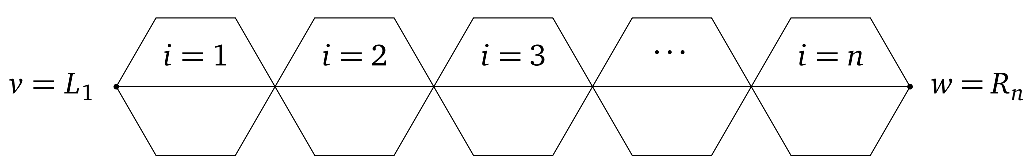


**Figure 7:** *Disjointness gadgets chained together.*

We conclude by noting that $|V(G)| = \Theta(n)$ and $|E(G)| = \Theta(n)$. As $|x| = |y| = \Theta(n)$, this completes the reduction and $\mathsf{TRAV}_2$ is at least as hard as DISJ. $\square$

Recall that in Claim 2, we showed that $D(\mathsf{TRAV}_2) = \Theta(n)$. With disjointness, we show that $R(\mathsf{TRAV}_2) = \Theta(n)$ too. This is because Claim 3 gives $R(\mathsf{TRAV}_2) = \Omega(n)$, since $R(\mathsf{DISJ}) = \Theta(n)$

(Rao and Yehudayoff 2020). Then, the trivial protocol where Alice sends her entire $G_A$ to Bob in $\Theta(n)$ bits works even in the public coin randomness model. Together this gives $R(\mathsf{TRAV}_2) = \Theta(n)$. In other words, we cannot do much better even with bounded error.

## 2.3 Specialized Two-Party Bounds

The previous proofs show that $\mathsf{TRAV}_2$ is maximally complicated in terms of deterministic communication complexity and public randomized communication complexity. A natural follow up question is to ask if we can do better. That is, given some additional constraints, can we solve the problem with reduced communication complexity?

**Definition 3.** *Define the* $\mathsf{CTRAV}_2$ *problem as a* $\mathsf{TRAV}_2$ *problem where* $G_A$ *and* $G_B$ *are guaranteed to be connected subgraphs of* $G$.

**Claim 4.** $\mathsf{CTRAV}_2$ *is at least as hard as* $\mathsf{DISJ}$. *Thus,* $D(\mathsf{CTRAV}_2) = \Theta(n)$ *and* $R(\mathsf{CTRAV}_2) = \Theta(n)$.

*Proof.* Let $x, y$ be inputs of Alice and Bob to the $\mathsf{DISJ}$ problem. We construct a graph $G$ as follows: Give ownership of $v$ and all vertices labelled $a_i$ and $i$ if $x_i = 1$, for $i \in [n]$. Similarly for $w$, $G_B$, and $y$. Then if there is a path from $v$ to $w$, this means $\exists i \in [n]$ such that $x_i = y_i = 1$. This means that $S_A \cap S_B \neq \emptyset$. On the other hand, if there is no path from $v$ to $w$, then for all $i \in [n]$, either $x_i = 0$ or $y_i = 0$.

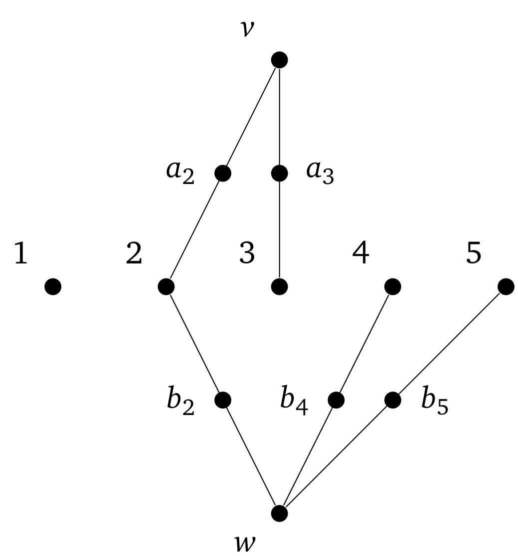


**Figure 8:** *A* $\mathsf{DISJ}$ *embedding into* $\mathsf{CTRAV}$ *with* $n = 5$. *There is a path from* $v$ *to* $w$ *if and only if* $\exists i \in [n]$ *with* $x_i = y_i = 1$.

We note that $|V(G_A)| = |V(G_B)| = |x| = |y| = \Theta(n)$. Furthermore, $G_A$ and $G_B$ are connected subgraphs of $G$. This completes the reduction. $\square$

Informally, this tells us that even when $G_A$ and $G_B$ are connected subgraphs, we do not immediately get better bounds and the problem does not become significantly easier. We now consider a sparser case:

**Definition 4.** *Define* $\mathsf{STRAV}_2$ *as a* $\mathsf{TRAV}_2$ *problem where* $E(G) = O(n)$, $E(G_A) = O(n)$, *and* $E(G_B) = O(n)$. *In other words, the graph* $G$ *is sparse and* $G_A$, $G_B$ *are sparse too.*

**Claim 5.** $\mathsf{STRAV}_2$ *is at least as hard as* $\mathsf{DISJ}$. *Thus,* $D(\mathsf{STRAV}_2) = \Theta(n)$ *and* $R(\mathsf{STRAV}_2) = \Theta(n)$.

*Proof.* We reuse the reduction from $\mathsf{DISJ}$ to $\mathsf{TRAV}_2$ in Claim 3. In that construction, $G$ is a chain of $n$ disjointness gadgets, so $|E(G)| = \Theta(n)$. Moreover, each player's subgraph consists of the gadget boundary vertices and one branch choice per gadget, which also gives $|E(G_A)| = \Theta(n)$ and $|E(G_B)| = \Theta(n)$.

Hence every instance produced by that reduction already satisfies the sparsity constraints in the definition of $\mathsf{STRAV}_2$. The reduction still preserves correctness: $\mathsf{DISJ}(x, y) = 1$ iff there is a $v$ to $w$ path in $G_A \cup G_B$. Therefore $\mathsf{STRAV}_2$ is at least as hard as $\mathsf{DISJ}$. □

Informally, this tells us that weak edge sparsity of $G$, $G_A$, and $G_B$ does not add nor remove significant communication cost. Combined with Claim 4, this may imply that the local structure of $G$, $G_A$, and $G_B$, namely, how sparse or connected the edges corresponding to the owned vertices are, do not have a significant impact on the deterministic and pubic randomized communication complexity of the $\mathsf{TRAV}_2$ problem.

# 3 The Multi-Party Traversal Problem

In Section 2, we analyzed $\mathsf{TRAV}_2$, the two-party traversal problem. We now study the $k$-party version, denoted $\mathsf{TRAV}_k$, with $k \geq 2$.

**Definition 5.** *Let* $G = (V, E)$ *be an undirected graph with designated start and end vertices* $v, w \in V$, *where* $|V| = n$. *There are* $k$ *players, and player* $i \in [k]$ *owns a vertex subset* $V_i \subseteq V$. *The edge set owned player* $i$ *is* $E_i = \{(a, b) \in E \mid \{u, v\} \cap V_i \neq \emptyset\}$. *Thus player* $i$ *owns a subgraph* $G_i = (V_i, E_i)$. *Given a fixed* $G$, *start vertex* $v$, *and end vertex* $w$ *known to all players, an instance of* $\mathsf{TRAV}_k$ *is defined as follows:*

$$\mathsf{TRAV}_k(G_1, \ldots, G_k) : \bigtimes_{i=1}^{k} \{0,1\}^n \to \{0,1\} \tag{6}$$

$$\mathsf{TRAV}_k(G_1, \ldots, G_k) = \begin{cases} 1 & \text{if there is a path from } v \text{ to } w \text{ in } \bigcup_{i=1}^k G_i, \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

**Definition 6.** *The* $k$*-party disjointness problem* $\mathsf{DISJ}_k$ *is defined as follows: Each player* $i \in [k]$

*has input $x^{(i)} \in \{0,1\}^n$, interpreted as a subset $S_i = \{j \in [n] \mid x^{(i)}_j = 1\} \subseteq [n]$. Define*

$$\mathsf{DISJ}_k(x^{(1)}, \ldots, x^{(k)}) : \bigtimes_{i=1}^{k} \{0,1\}^n \to \{0,1\} \tag{8}$$

$$\mathsf{DISJ}_k(x^{(1)}, \ldots, x^{(k)}) = \begin{cases} 1 & \bigcap_{i=1}^k S_i = \emptyset, \\ 0 & \bigcap_{i=1}^k S_i \neq \emptyset. \end{cases} \tag{9}$$

It is well known that $D(\mathsf{DISJ}_k) = \Omega(nk)$ and $R(\mathsf{DISJ}_k) = \Omega(nk)$ (Braverman et al. 2013).

**Claim 6.** *$\mathsf{TRAV}_k$ is at least as hard as $\mathsf{DISJ}_k$. Furthermore, $D(\mathsf{TRAV}_k) = \Theta(nk)$ and $R(\mathsf{TRAV}_k) = \Theta(nk)$.*

*Proof.* Let $(x^{(1)}, \ldots, x^{(k)})$ be an input to $\mathsf{DISJ}_k$ with $x^{(i)} \in \{0,1\}^n$. The high level idea is similar to Claim 3: we build a $\mathsf{TRAV}_k$ instance from a chain of $n$ multi-party disjointness gadgets.

**Multi-party disjointness gadget.** For each index $j \in [n]$, create vertices $L_j$, $R_j$, and $M_j$. All players own $L_j$ and $R_j$. Furthermore, player $i$ owns $M_j$ if and only if $x^{(i)}_j = 0$. Then there is a valid path from $L_j$ to $R_j$ if and only if at least one player has $x^{(i)}_j = 0$. Equivalently, there is no such path if and only if all players have $x^{(i)}_j = 1$.

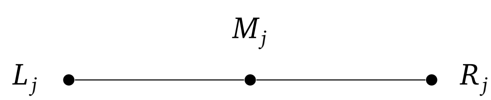


**Figure 9:** *Gadget for embedding $\mathsf{DISJ}_k$. A valid path $L_j$ to $R_j$ exists if and only if $\exists i \in [k]$ with $x^{(i)}_j = 0$.*

**Reduction to $\mathsf{TRAV}_k$.** We construct a graph $G$ by chaining $n$ copies of the multi-party disjointness gadget, connecting $R_l = L_{l+1}$ for all $l \in [n-1]$. Set the start vertex $v = L_1$ and end vertex $w = R_n$. Then there is a path from $v$ to $w$ in $\bigcup_{i=1}^k G_i$ if and only if every gadget is traversable (that is, there is a path $L_j$ to $R_j$ for all $j \in [n]$). This means that for every $j$, there is some player with $x^{(i)}_j = 0$, so $\bigcap_{i=1}^k S_i = \emptyset$. In other words, a path from $v$ to $w$ witnesses that every element in $[n]$ is missing from at least one subset $S_i$.

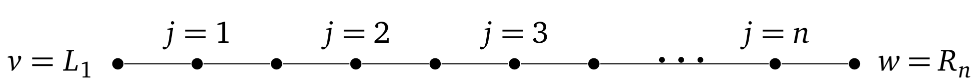


**Figure 10:** *$n$ multi-party disjointness gadgets chained together.*

We conclude the proof by noting that the construction uses $\Theta(n)$ vertices and edges. Hence $D(\mathsf{TRAV}_k) = \Omega(nk)$ and $R(\mathsf{TRAV}_k) = \Omega(nk)$.

For upper bounds, consider the trivial protocol where players $2, \ldots, k$ send their input to player 1 in $(k-1)n = \Theta(nk)$ bits. Then Player 1 can compute the answer. This trivial protocol works in both the deterministic and randomized setting. This gives $D(\mathsf{TRAV}_k) = \Theta(nk)$ and $R(\mathsf{TRAV}_k) = \Theta(nk)$. $\square$

Informally, this tells us that adding more players does not make traversal an easier problem in terms of deterministic and public randomized communication complexity.

# 4 Closing Remarks

In this paper, we explored the communication complexity of multiparty graph traversal as defined in the TRAV problem and its variants. By exploring embeddings of equality and disjointness, we show TRAV is maximally hard under a variety of conditions in the deterministic and public randomized communication complexity models. In other words, given the most general $\mathsf{TRAV}_k$ for $k \geq 2$, one cannot asymptotically do better than the trivial solution of sending all inputs to one player.

This has a variety of implications for algorithm, system, and database designers. If such specialists were to implement algorithms solving $\mathsf{TRAV}_k$ in it its most general form, we have proven that there are no "communication tricks" that can be used to get or even approximate the right answer. More formally, there does not exist an algorithm that computes $\mathsf{TRAV}_k$ asymptotically better than the trivial protocol that attains at most a constant bounded error against every possible input.

In particular, we note that the trivial protocol can be implemented by having one player act as the "coordinator," receiving and processing everyone's inputs. In terms of minimizing bits communicated, this means systems that solve problems like $\mathsf{TRAV}_k$ may benefit from having a single, optimized server as opposed to distributed computing. Of course, there are practical, economical, and privacy reasons that may complicate the choice of implementation and the priority of minimizing communication in practice.

## 4.1 Future Work

Though we have shown that $\mathsf{TRAV}_k$ and some of its variants are maximally hard under the deterministic and randomized model, it is entirely possible that specialized instances of $\mathsf{TRAV}_k$ give rise to clever algorithms and lower bounds that have significantly less communication complexity. Below, we propose some such variants, which due to practical constraints we cannot examine in this paper.

1. **Bounded-length path**. Consider a $\mathsf{TRAV}_2$ problem where we know that if there is a path from $v$ to $w$ in $G$, then this path is of length at most $l = O(1)$. Does this give rise to a significantly lower communication complexity? What about for $\mathsf{TRAV}_k$?
2. **Single-connected component**. We noted that $\mathsf{CTRAV}_2$ is also maximally complicated in the deterministic and public coin model. Notice that $G_A$ and $G_B$ in $\mathsf{CTRAV}_2$

are connected components. If we guarantee in $\mathsf{TRAV}_k$ that each $G_i$ is a single connected component, can we achieve lower communication complexity?

3. **Bounded-owners**. Suppose it is known that each vertex in $\mathsf{TRAV}_k$ can be owned by at most $p = O(1)$ players. For example, if $p = 1$, each vertex can be owned by at most 1 player. Does this give rise to improved communication lower and upper bounds?

It may also be worthwhile to investigate very specific instances of TRAV based on real-world networks and graphs. For example, perhaps a combination of the above variants models a specific, real-world problem that gives rise to clever algorithms with non-maximal communication complexity. That said, one benefit of studying the most general cases is the ability for the results to be applicable to a wide range of problems, scenarios, and applications.

# 5 References

Braverman, Mark et al. (2013). "Tight Bounds for Set Disjointness in the Message Passing Model". In: *CoRR* abs/1305.4696. arXiv: 1305.4696. URL: http://arxiv.org/abs/1305.4696.

Rao, Anup and Amir Yehudayoff (2020). *Communication Complexity: and Applications*. Cambridge University Press.